Casey Barton

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PROFILE

I come at this role from both ends of the homebuilding workflow. My working background is home inspection — permits, plan sets, elevations, options, site walkthroughs, materials, and punch lists on real jobsites — so I have operated the document and jobsite workflows Higharc turns into actionable data. The posting names residential-construction fluency as a major plus; plan reading is my native language, not a credential I studied.

My engineering practice is applied AI systems: agent-enabled search and RAG over document libraries, evaluation harnesses, failure-mode recovery, and trace intelligence across agent and retrieval pipelines, with **65** authored MCP connectors and typed contracts governing every tool boundary. Three years as a founder running an evidence-grade agent control plane from Honolulu: embedded with operators, scoping ambiguous work, prototyping rapidly, hardening what carries load. CAD/BIM fluency is reading-level exposure, honestly stated.

CORE COMPETENCIES

Construction domain: home-inspection background — permits · plan sets & elevations · plan options / variation management · site walkthroughs · materials assessment · punch lists · photo / site documentation · jobsite coordination workflows

Deployed delivery: embedded delivery posture · customer discovery · ambiguous-scope scoping · rapid prototyping → production · field friction → product requirements · technical-consulting posture · trusted builder relationships

AI & Evaluation: agent systems · RAG pipelines · agent-enabled natural-language search · evaluation harnesses · golden datasets · behavioral assertions · LLM-as-a-judge with written rubrics · quality scoring · trace intelligence · MCP servers & connectors · function calling · structured outputs · guardrails · fail-closed recovery

Production: Python (primary) · TypeScript · SQL · Bash · REST APIs · JSON Schema / Pydantic-style validation · GitHub Actions (CI/CD) · Postgres · per-task latency & token-cost instrumentation · retry logic · observability & run logging · documentation engineers actually read

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Deliver like a Forward Deployed Engineer: embed on ambiguous workflows, scope the unclear problem with operators, prototype rapidly, harden what carries load — the founding-FDE motion of turning plans and documents into actionable production data.
- Apply construction-domain fluency as a first-order asset: permits, plan sets, elevations, options, punch lists, and site documentation — firsthand knowledge of how builders manage plan variation, brought to data and AI pipelines that serve those workflows.

- Build agent-enabled search and RAG over document libraries: natural-language retrieval over plan-like corpora, retrieval quality gated by eval harnesses with written rubrics — quality measured before it ships, not argued after.
- Design agent systems for failure: instrument the failure modes found in production — stuck loops, bad retrievals, tool misuse, context blowouts — and build the catchers: replay loops, fail-closed gates, behavioral assertions, recovery restricted to evidence declared before reassessment.
- Turn field friction into product direction: trace and run evidence feeding prompt, retrieval, and configuration iteration — embedded-delivery feedback loops, documented and auditable.
- Run an evidence-grade agent control plane across a **1,395-repository** estate: typed handoff contracts, guardrails, recoverable runs — backed by a **989-test** suite, **62/62** verification checks green.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — evaluation-gated campaign engine

- Attributable campaign engine with immutable evidence models and fail-closed **GO / NO-GO** decisions — the eval gate decides what carries load.
- Judge criteria with written rubrics behind every automated quality decision — the quality discipline retrieval-heavy AI products depend on.

Construction-domain edge

- Home-inspection background: permits, plan sets, elevations, options, site walkthroughs, materials, punch lists — the workflows Higharc's product serves, operated firsthand.
- CAD/BIM exposure at the reading level only — no production authoring claims. Plan-set fluency, honestly stated.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

Construction domain: plan reading · permits · elevations & options · punch lists · site documentation · materials assessment · jobsite coordination

Deployed engineering: discovery & scoping · rapid prototyping · embedded delivery · field-to-product feedback · handover discipline · technical writing

AI / Evaluation: agent systems · RAG · eval harnesses · golden datasets · LLM-as-a-judge · trace intelligence · MCP · structured outputs · fail-closed recovery

Evidence boundary: portfolio repositories are independent work. No enterprise deployment logos claimed; customer-facing delivery is founder/operator plus inspection client work (three years), honestly stated — the 5+ year line is a stretch, flagged. No Higharc-internal production claims. GitHub is the forge of record.